



Emanuele Andarolo

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About me

I'm Emanuele, a Computer Science and Engineering student with a passion for machine learning, cybersecurity, and edge AI systems. I describe myself as an eclectic and curious person who adapts seamlessly to any context. I like to explore different ideas and I eagerly embrace new challenges and opportunities.

Education

Politecnico di Milano

Sept 2023 – Present

Master's Degree (M.Sc.) in Computer Science and Engineering

Thesis: "ML-Based Fraud Detection for Public Transport Ticketing Systems" **Università di Catania**

Sept 2018 – Feb 2023

Bachelor's Degree (B.Sc.) in Computer Science and Engineering

graduated with the highest score (110/110).

Thesis: "Cost/Performance Evaluation of Different Data Quantization Levels in Hardware Accelerators for CNN"

I.T.T. "Ettore Majorana"

Sept 2014 – Jun 2018

High School Diploma as an Industrial Computer Technician specialized in Telecommunications

graduated with the highest grade (100/100).

Summer Schools & Courses

ETH Zürich

Aug 2026

Summer School – "Co-creating Transformative Knowledge: Inter- and Transdisciplinary Research in Practice"

Completed. Selected through a competitive process; scholarship awarded by the ENHANCE alliance.

Publications

- E. Andarolo, R. Russo, F. Dercole, "Real-time AI-powered moving pattern recognition from low-cost accelerometer and gyroscope data," *7th International Interdisciplinary Chaos Symposium on Chaos and Complex Systems (SCCS2026)*, Athens, Greece, June 2026. (Accepted, Springer proceedings)

Projects

- **ML-Based Fraud Detection for Public Transport (MSc Thesis):** Developing a machine learning pipeline for detecting forged and fraudulent tickets in public transportation systems, working on two datasets (validation tickets and sold tickets). The work involves anomaly detection, threshold calibration across cross-validation and full-data models, and a split-abuse detection pipeline for season passes.
- **RLC Language Server (Open Source Contribution):** Contributed improvements to the LSP (Language Server Protocol) tooling of the RLC language (rl-language/rlc) in C++/MLIR, supervised by author. Implemented scope-aware code completion, completion ranking via `sortText`, document symbols (Outline / Go to Symbol), and a configurable stdlib path. Developed a coverage-guided fuzzer (libFuzzer) that exercises the LSP server, and submitted the work as a pull request.
- **Leonardo Project at NecstLab:** Worked on an AI-driven well-being app designed to foster empathetic connections with users through voice-based interactions. This project won the hackathon *How To Hack The NECSTCamp*, leading to an invitation to the NECST Group Conference (NGC14) in San Francisco, where we presented it to major tech companies and startups.
- **TOP program:** Participated as a mentor in the innovative Online Tutoring Program (TOP), supported by Fondazione Cariplo in collaboration with Bocconi University, Harvard University, and Bicocca University. Provided targeted academic support through online tutoring sessions to middle school students.
- **BlockVerse Project:** Contributed to the BlockVerse research project, developing a user-friendly dashboard integrated with Katena, a blockchain application management framework compatible with Ethereum. Implemented topology visualization, management, and YAML export functionalities. Managed backend interactions and user authentication utilizing Firebase.
- **Sensor Data Collection Application:** Developed a mobile application in Flutter for collecting real-time sensor data (magnetometer, accelerometer, gyroscope) with Firebase cloud storage and visualization.
- **Deployment of CNN and Custom ViT on STM32N6 Board with NPU:** Deployed CNNs and customized Vision Transformer (ViT) models on the STM32N6570-DK board leveraging its Neural Processing Unit (NPU), with model quantization for real-time inference.
- **Reflectance Spectroscopy-Based Waste Recognition Prototype (E-Trash):** Developed a low-cost reflectance spectroscopy module for automatic waste classification in collaboration with the startup E-Trash and Politecnico di Milano.

Experiences

Explorer Tech (Computer repair shop)	<i>Sep 2017 – Oct 2017</i>
Computer maintenance and troubleshooting; deployed server environments (infrastructure and networking).	
Nagual Informatica (Computer repair shop)	<i>Jul 2015 – Aug 2015</i>
Hardware/software maintenance and technical support; installed and configured Oracle-based server systems.	

Skills

Programming Languages: C + + , C, Java, Python, C#, SQL, JavaScript, PHP, Flutter, HTML, CSS, VHDL, Arduino Blink, LaTeX, Expo, Node.js
Technologies: Laravel, MS Office, XAMPP, Arduino Boards, Raspberry Pi, Flask, Elastic Search, Kibana, MongoDB, DynamoDB, Neo4J, Vulkan
ML & AI: PyTorch, TensorFlow Lite, ONNX Runtime, GRU/LSTM, CNNs, Vision Transformers, Anomaly Detection, Edge AI
Spoken Languages: Italian (Native Speaker), English (Cambridge First, Score: 173), Spanish (Good level)

Volunteering

Blood Donor AVIS	<i>Nov 2022 – Present</i>
Field: Health	
Volunteering at IEEE IPDPS 2024 Conference	<i>May 2024 – Jun 2024</i>
IEEE Computer Society	
• Served as a volunteer on the organizing team for the 38th IEEE International Parallel and Distributed Processing Symposium in San Francisco. • Contributed to conference logistics, attendee support, and event management.	